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# Assignment 1: Logic and Set Theory

## Important Rules

- Each question is worth 10 points.
- You may discuss the problems with classmates.
- However, the work you submit must be entirely your own.
- You are not allowed to use AI tools (including ChatGPT or similar systems) to generate answers.
- Any submitted solution must reflect your own understanding and reasoning.

If you discuss ideas with others, make sure you can independently explain every step of your final answer.

## 1. Recursive Definition of a Set

Consider the set:

$$S = \{2,6,7,21,22,66,67,201,202, \dots\}$$

(a) Describe the pattern in words. Starting from 2, the pattern is to triple it, then add 1, then triple, then add 1, repeat infinitely.

(b) Define the set recursively using the three-step process:

| | |
|---|---|
| 1. Basis step | $5 \in A$ |
| 2. Recursive step | $\forall a \in A, 2a \in B$<br>$\forall b \in B, b + 3 \in A$ |
| 3. Closure condition | $S = A \cup B$ |

Your definition must clearly specify the initial element(s), how new elements are generated, and that nothing else belongs to the set.

## 2. Propositional Logic

Let:

p: 'The device is on.'

q: 'The sensor is recording.'

r: 'The system is logging data.'

(a) Write the following statement in symbolic form: $\neg p \lor \neg q \Rightarrow \neg r$

'If the device is not on, or the sensor is not recording, then the system is not logging data.'

(b) Rewrite the statement using logical equivalences to simplify it as much as possible.

$$\neg(p \ \land \ q) \Rightarrow \neg r$$

**3. Predicate Logic and Quantifiers**

Let the domain be all students at the university.

S(x): 'x is a student'

M(x): 'x has taken a mathematics course'

(a) Express symbolically: 'Every student has taken a mathematics course.' $\forall x \in S(x), x \in M(x)$

(b) Write the negation of your statement using quantifiers. $\exists x \in S(x), x \notin M(x)$

(c) Explain in plain English what the negation means.

There exists a student that hasn't taken a mathematics course

**4. Sets and Operations**

Let:

$$A = \{ x \ in \ \mathbb{N} | \ x \ is \ even \ and \ x < 10 \}$$

$$B = \{ x \ in \ \mathbb{N} \ | \ x \ is \ a \ multiple \ of \ 3 \ and \ x < 10 \}$$

$$U = \{1,2,3,4,5,6,7,8,9,10\}$$

(a) List the elements of A and B. A = {2,4,6,8}, B = {3,6,9}

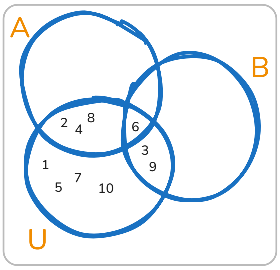
(b) Find: A ∪ B, A ∩ B, and A' (complement of A in U).

(c) Draw a Venn diagram representing A, B, and U.

(b)

$$A \cup B = \{2, 3, 4, 6, 8, 9\},$$
$$A \cap B = \{6\},$$
$$A' = \{1, 3, 5, 7, 9, 10\}$$

(c)




**5. Cartesian Product and Power Set**

Let

$$X = \{1,2\} \ and \ Y = \{a, b\}.$$

(a) Write out the Cartesian product $X \times Y$. = {(1,a), (2,a), (1,b), (2,b)}

(b) Find the power set $P(X)$. = {∅, {1}, {2}, {1,2}}

(c) How many elements does $P(X)$ have? Explain how you determined that number.

P(X) has 4 elements because $2^2 = 4$.